

Microphone Pre-Amplifier with Digital Output

FAN3852

Description

The FAN3852 integrates a pre-amplifier, LDO, and ADC that converts Electret Condenser Microphone (ECM) outputs to digital Pulse Density Modulation (PDM) data streams. The pre-amplifier accepts analog signals from the ECM and drives an over-sampled sigma delta Analog-to-Digital Converter (ADC) and outputs PDM data. The PDM digital audio has the advantage of noise rejection and easy interface to mobile handset processors.

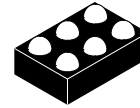
The FAN3852 features an integrated LDO and is powered from the system supply rails up to 3.63 V, with low power consumption of only 0.85 mW and less than 20 μ W in Power-Down Mode.

Features

- Optimized for Mobile Handset and Notebook PC Microphone Applications
- Accepts Input from Electret Condenser Microphones (ECM)
- Pulse Density Modulation (PDM) Output
- Standard 5-Wire Digital Interface
- Low Input Capacitance, High PSR, 20 kHz Pre-Amplifier
- Low-Power 1.5 μ A Sleep Mode
- Typical 420 μ A Supply Current
- SNR of 62 dB (A) for 16 dB Gain
- Total Harmonic Distortion 0.02%
- Input Clock Frequency Range of 1–4 MHz
- Integrated Low Drop-Out Regulator (LDO)
- Small 1.242 mm $\times$ 0.842 mm 6-Ball, 0.400 mm pitch standard WLCSP Package
- 1.5 kV HBM ESD on MIC Input

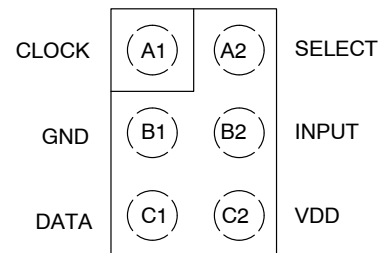
Typical Applications

- Electret Condenser Microphones with Digital Output
- Mobile Handset
- Headset Accessories
- Personal Computer (PC)



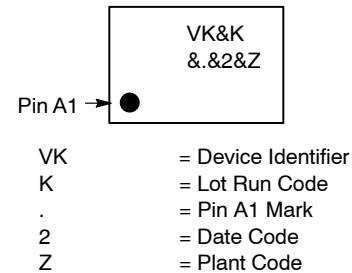
WLCSP-6
CASE 567TS

PIN CONFIGURATION



Top View

MARKING DIAGRAM



ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

FAN3852

ORDERING INFORMATION

Part Number	Operating Temperature Range	Package	Packing Method†
FAN3852UC16X	-40°C to 85°C	6 Ball, Wafer-Level Chip-Scale Package (WL CSP)	3000 Units/Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

INTERNAL BLOCK DIAGRAM

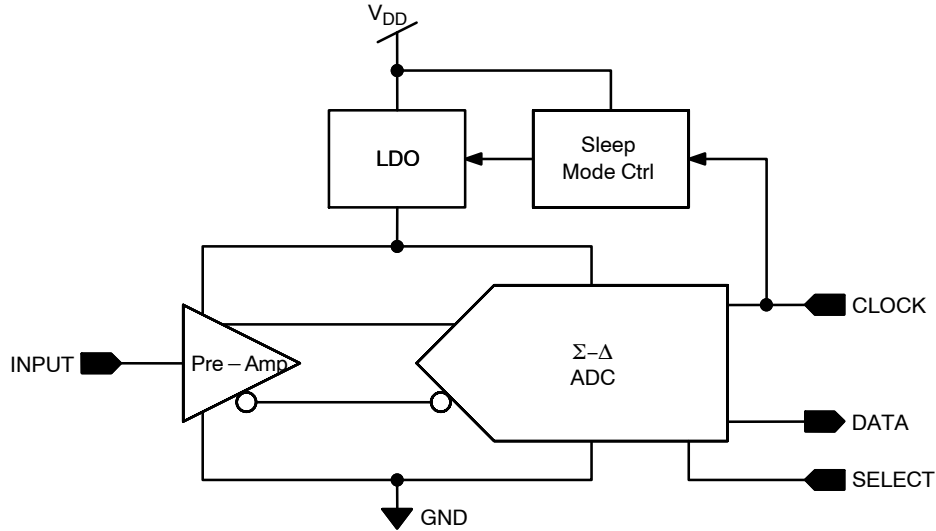


Figure 1. Block Diagram

Table 1. PIN DEFINITIONS

Pin #	Name	Type	Description
A1	CLOCK	Input	Clock Input
B1	GND	Input	Ground Pin
C1	DATA	Output	PDM Output – 1 Bit ADC
A2	SELECT	Input	Rising or Falling Clock Edge Select
B2	INPUT	Input	Microphone Input
C2	VDD	Input	Device Power Pin

Table 2. ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Min.	Max.	Unit
V_{DD}	DC Supply Voltage	-0.3	4.0	V
V_{IO}	Digital I/O	-0.3	$V_{DD} + 0.3$	V
	Microphone Input	-0.3	2.2	
ESD	Human Body Model, JESD22-A114, All Pins Except Microphone Input	±8		kV
	Human Body Model, JESD2-A114 – Microphone Input	±1.5		

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. This device is fabricated using CMOS technology and is therefore susceptible to damage from electrostatic discharges. Appropriate precautions must be taken during handling and storage of this device to prevent exposure to ESD.

Table 3. RELIABILITY INFORMATION

Symbol	Rating	Min.	Typ.	Max.	Unit
T_J	Junction Temperature			+150	°C
T_{STG}	Storage Temperature Range	-65		+125	°C
T_{RFLW}	Peak Reflow Temperature			+260	°C
θ_{JA}	Thermal Resistance, JEDEC Standard, Multilayer Test Boards, Still Air		90		°C/W

2. $T_A = 25^\circ\text{C}$ unless otherwise specified

Table 4. RECOMMENDED OPERATING CONDITIONS

Symbol	Rating	Min.	Typ.	Max.	Unit
T_A	Operating Temperature Range	-40		+85	°C
V_{DD}	Supply Voltage Range	1.64	1.80	3.63	V
T_{RF-CLK}	Clock Rise and Fall Time			10	ns

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

Table 5. DEVICE SPECIFIC ELECTRICAL CHARACTERISTICS

Symbol	Value	FAN3852UC16X			Unit
		Min.	Typ.	Max.	
SNR	Signal-to-Noise Ratio $f_{IN} = 1 \text{ kHz}$ (1 Pa), A-Weighted		62		dB (A)
e_N	Total Input RMS Noise 20 Hz to 20 kHz, A-Weighted		5.74	6.80	μV_{RMS}
V_{IN}	Maximum Input Signal $f_{IN} = 1 \text{ kHz}$, THD + N < 10%, Level = 0 V			448	mV _{PP}

3. Guaranteed by characterization and/or design. Not production tested.

Table 6. ELECTRICAL CHARACTERISTICS

Unless otherwise specified, all limits are guaranteed for $T_A = 25^\circ\text{C}$, $V_{DD} = 1.8 \text{ V}$, $V_{IN} = 94 \text{ dB (SPL)}$ and $f_{CLK} = 2.4 \text{ MHz}$.
Duty Cycle = 50% and $C_{MIC} = 15 \text{ pF}$

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
V_{DD}	Supply Voltage Range		1.64	1.80	3.63	V
I_{DD}	Supply Current	INPUT = AC Coupled to GND, CLOCK = On, No Load		420		μA
I_{SLEEP}	Sleep Mode Current	$f_{CLK} = \text{GND}$		1.50	8.0	μA
PSR	Power Supply Rejection (Note 5)	INPUT = AC Coupled to GND, Test Signal on $V_{DD} = 217 \text{ Hz}$, Square Wave and Broadband Noise (Note 4), Both 100 mV _{P-P}		-74		dBFS
IN_{NOM}	Nominal Sensitivity (Note 6)	INPUT = 94 dB SPL (1 Pa)		-26		dBFS
THD	Total Harmonic Distortion (Note 7)	$f_{IN} = 1 \text{ KHz}$, INPUT = -26 dBFS		0.02	0.20	%
THD+N	THD and Noise (Note 5)	$50 \text{ Hz} \leq f_{IN} \leq 1 \text{ kHz}$, INPUT = -20 dBFS		0.2	1.0	%
		$f_{IN} = 1 \text{ KHz}$, INPUT = -5 dBFS		1.0	5.0	
		$f_{IN} = 1 \text{ KHz}$, INPUT = 0 dBFS		5.0	10.0	
C_{IN}	Input Capacitance (Note 8)	INPUT		1.3		pF
R_{IN}	Input Resistance (Note 8)	INPUT	>10			G Ω
V_{IL}	CLOCK & SELECT Input Logic LOW Level				0.3	V



Table 6. ELECTRICAL CHARACTERISTICS (continued)

Unless otherwise specified, all limits are guaranteed for $T_A = 25^\circ\text{C}$, $V_{DD} = 1.8\text{ V}$, $V_{IN} = 94\text{ dB (SPL)}$ and $f_{CLK} = 2.4\text{ MHz}$.
Duty Cycle = 50% and $C_{MIC} = 15\text{ pF}$

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
V_{IH}	CLOCK & SELECT Input Logic HIGH Level		1.5		$V_{DD}+0.3$	V
V_{OL}	Data Output Logic LOW Level				$0.35 \cdot V_{DD}$	V
V_{OH}	Data Output Logic HIGH Level		$0.65 \cdot V_{DD}$			V
V_{OUT}	Acoustic Overload Point (Note 8)	THD+N < 10%	120			dB SPL
t_A	Time from CLOCK Transition to Data becoming Valid	On Falling Edge of CLOCK, SELECT = GND, $C_{LOAD} = 15\text{ pF}$	18	43		ns
t_B	Time from CLOCK Transition to Data becoming HIGH-Z	On Rising Edge of CLOCK, SELECT = GND, $C_{LOAD} = 15\text{ pF}$	0	5	16	ns
t_A	Time from CLOCK Transition to Data becoming Valid	On Rising Edge of CLOCK, SELECT = V_{DD} , $C_{LOAD} = 15\text{ pF}$	18	58		ns
t_B	Time from CLOCK Transition to Data becoming HIGH-Z	On Falling Edge of CLOCK, SELECT = V_{DD} , $C_{LOAD} = 15\text{ pF}$	0	5	16	ns
f_{CLK}	Input CLOCK Frequency (Note 9)	Active Mode	1.0	2.4	4.0	MHz
CLK_{dc}	CLOCK Duty Cycle (Note 5)		40	50	60	%
t_{WAKEUP}	Wake-Up Time (Note 10)	$f_{CLK} = 2.4\text{ MHz}$		0.35	2.00	ms
$t_{FALLASLEAP}$	Fall-Asleep Time (Note 11)	$f_{CLK} = 2.4\text{ MHz}$	0	0.01	1.00	ms
C_{LOAD}	Load Capacitance on Data				100	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

4. Pseudo-random noise with triangular probability density function. Bandwidth up to 10 MHz.

5. Guaranteed by characterization. Not production tested.

6. Assuming that 120 dB (SPL) is mapped to 0 dBFS.

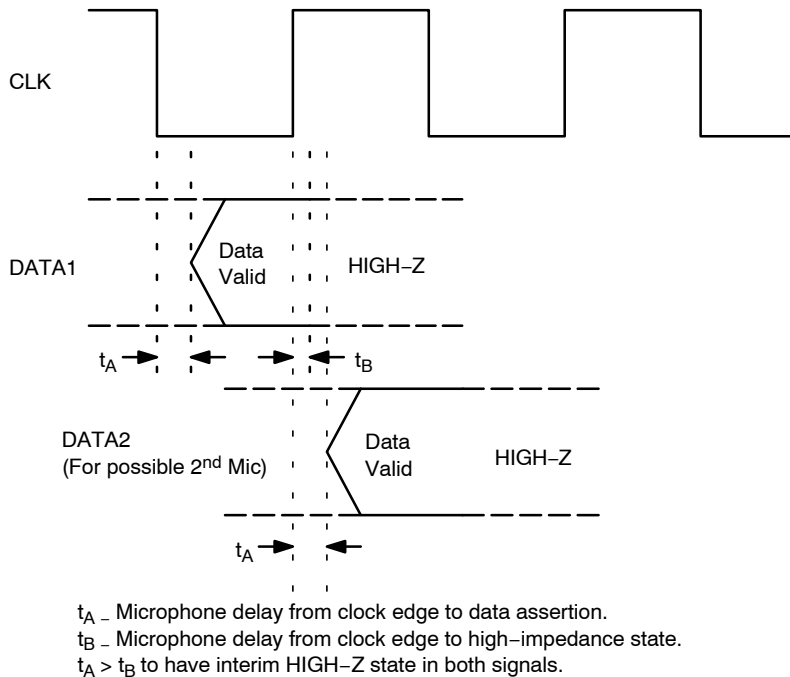
7. Assuming an input of -42 dBV (maps to -26 dBFS, 94 dB SPL).

8. Guaranteed by design. Not production tested.

9. All parameters are tested at 2.4 MHz. Frequency range guaranteed by characterization.

10. Device wakes up when $f_{CLK} \geq 300\text{ kHz}$.

11. Device falls asleep when $f_{CLK} \leq 70\text{ kHz}$.

**Figure 2. Interface Timing**

TYPICAL PERFORMANCE CHARACTERISTICS

Unless otherwise specified, all limits are guaranteed for $T_A = 25^\circ\text{C}$, $V_{DD} = 1.8\text{ V}$, $V_{IN} = 94\text{ dB (SPL)}$, $f_{CLK} = 2.4\text{ MHz}$ and duty Cycle = 50%.

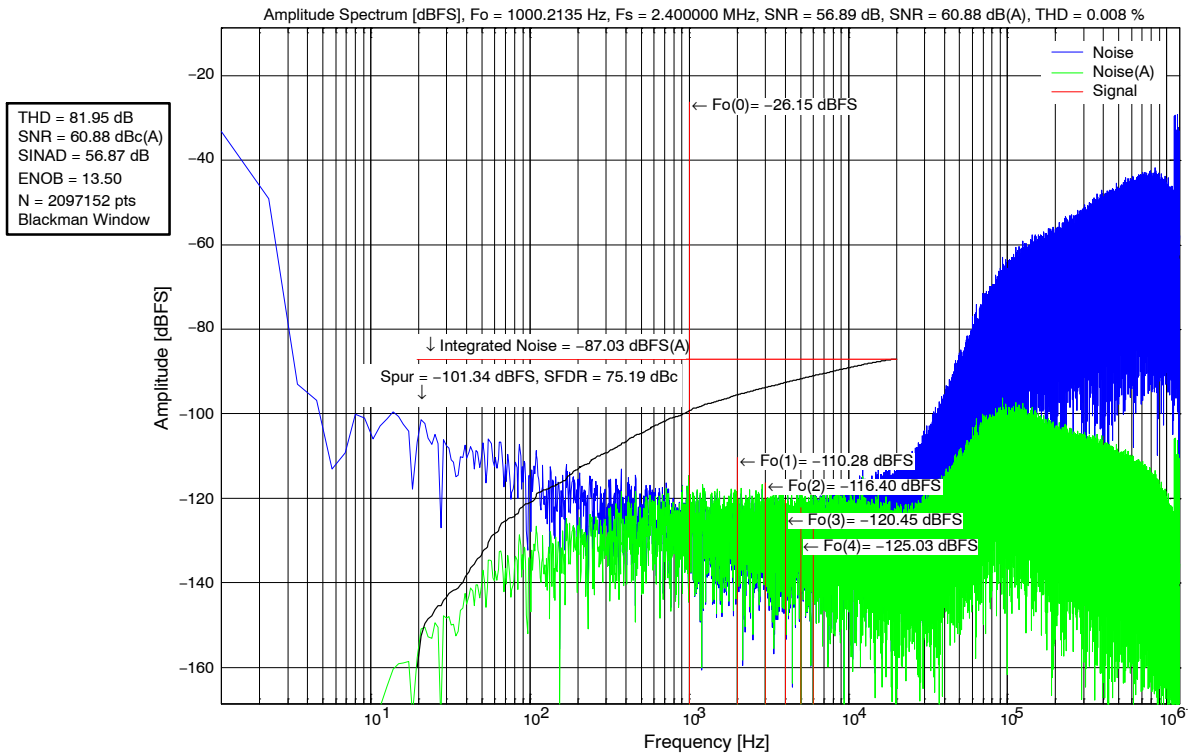


Figure 3. Noise vs. Frequency

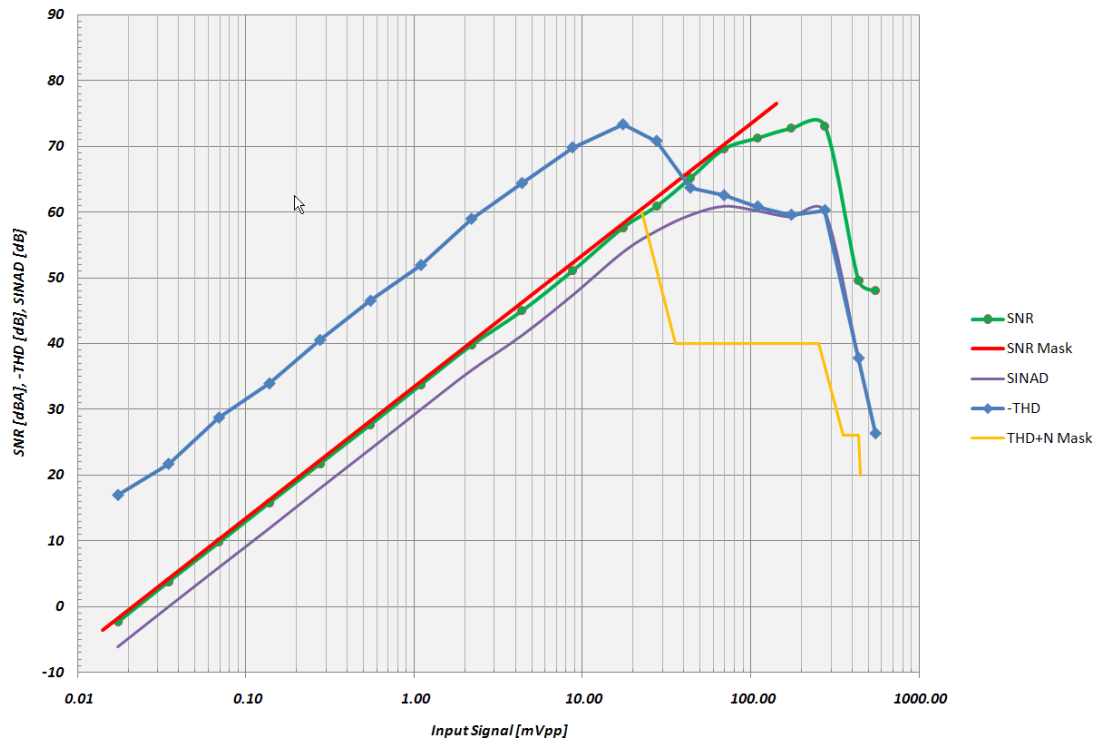


Figure 4. THD, SINAD, and SNR vs. Input Amplitude

FAN3852

TYPICAL PERFORMANCE CHARACTERISTICS (continued)

Unless otherwise specified, all limits are guaranteed for $T_A = 25^\circ\text{C}$, $V_{DD} = 1.8\text{ V}$, $V_{IN} = 94\text{ dB (SPL)}$, $f_{CLK} = 2.4\text{ MHz}$ and duty Cycle = 50%.

SNR, - THD, SINAD vs Output Level - FAN3850A-OPT1

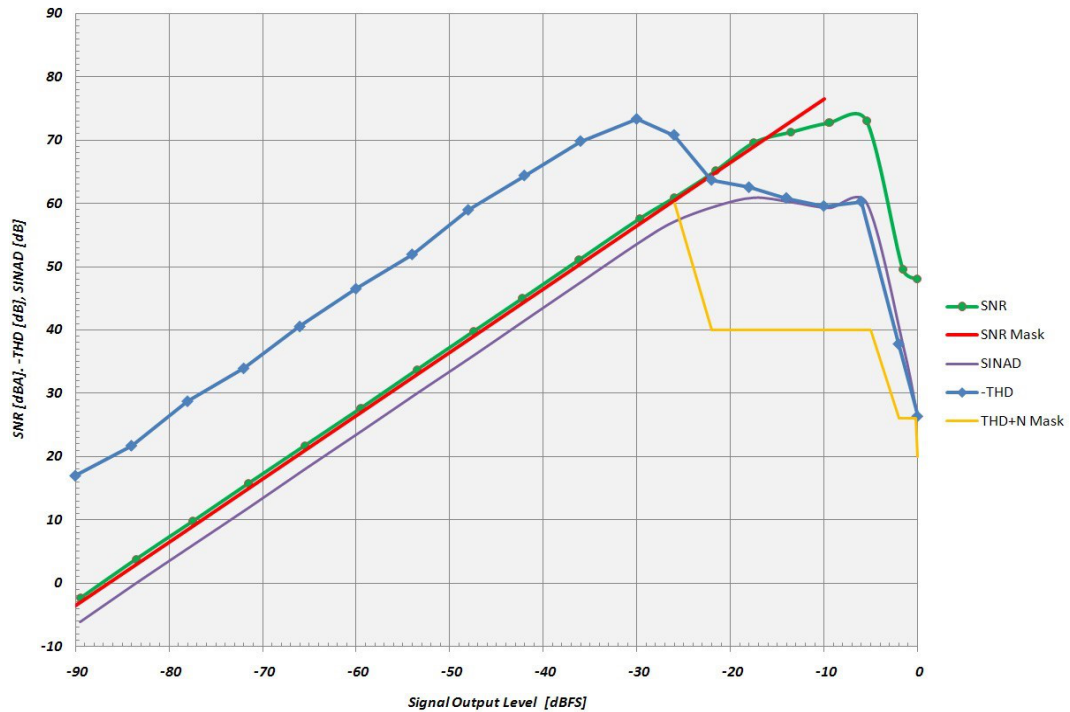


Figure 5. THD, SINAD and SNR vs. Output Level

Temp (°C)	Delta (dB)
-40	0.1971
-30	0.1644
-20	0.1260
-10	0.0954
0	0.0657
10	0.0359
20	0.0139
25	0.0000
30	-0.0097
40	-0.0344
50	-0.0514
60	-0.0739
70	-0.0998
80	-0.1183
85	-0.1271

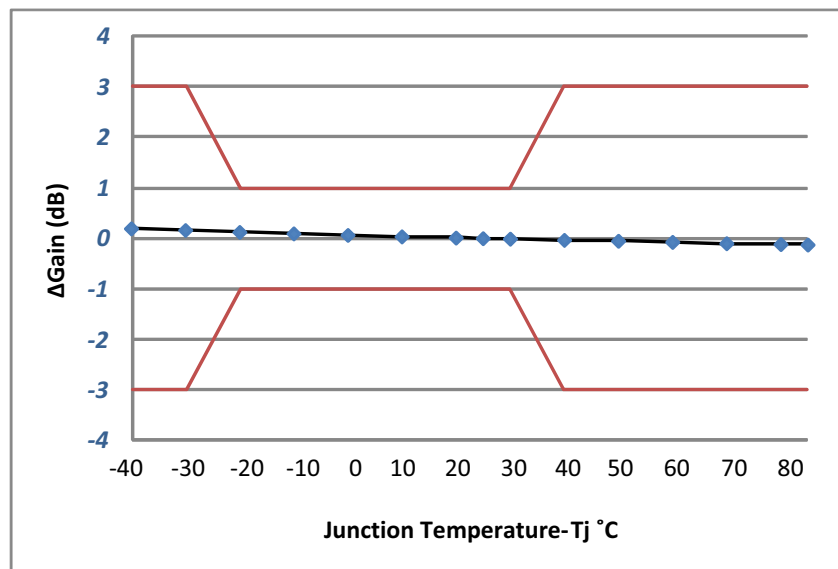


Figure 6. Δ Gain vs. Temperature (Nominal Temperature = 25°C)

APPLICATIONS INFORMATION

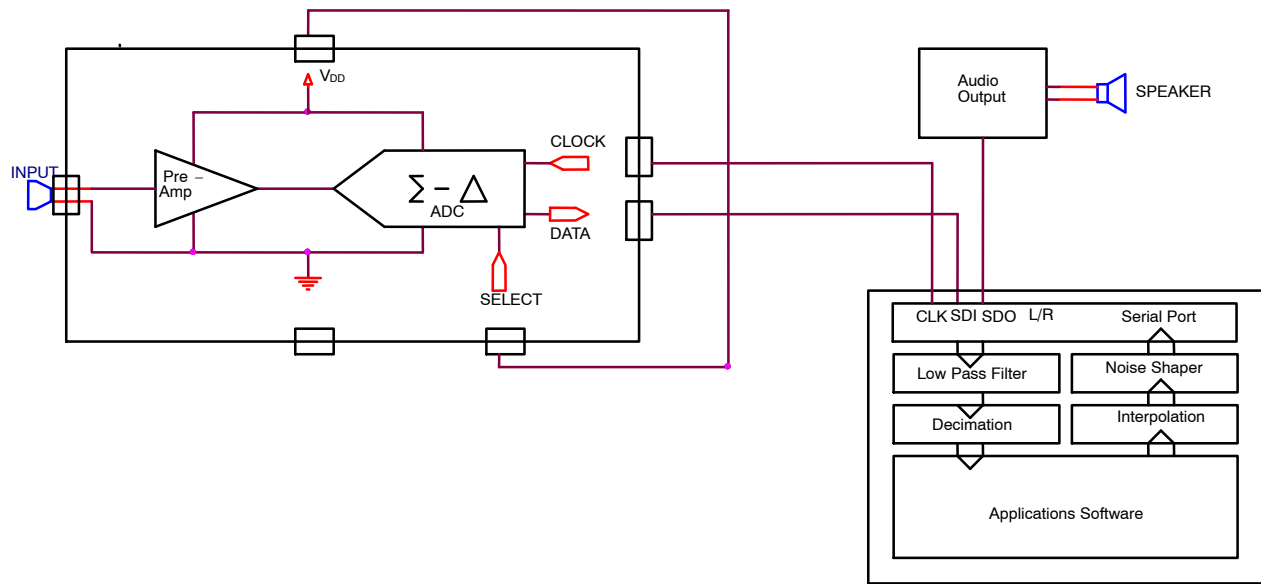


Figure 7. Mono Microphone Application Circuit

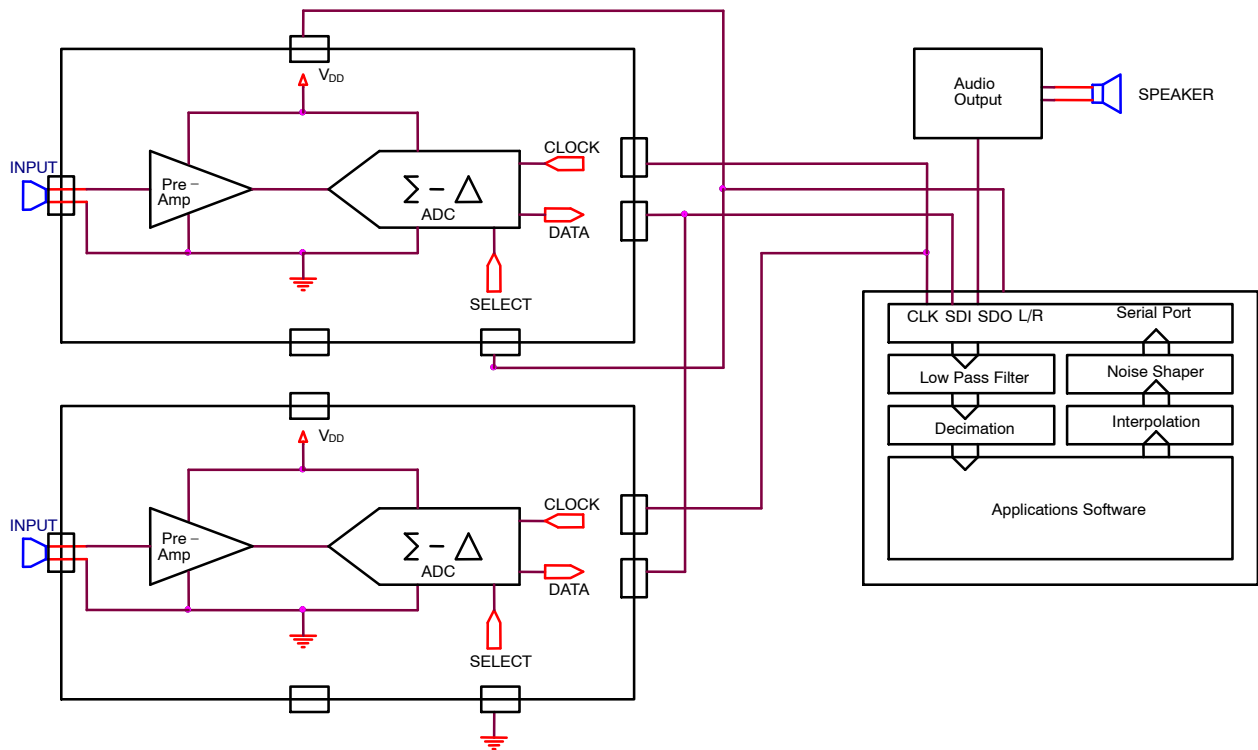


Figure 8. Stereo Microphone Application Circuit

FAN3852

APPLICATIONS INFORMATION (continued)

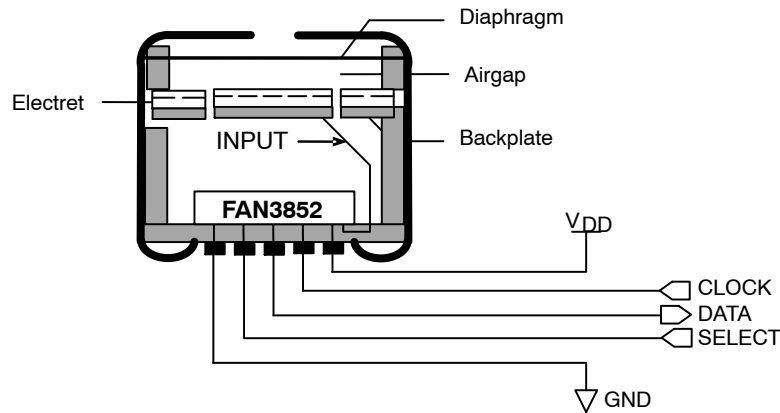


Figure 9. MIC Element Drawing

A 0.1 μF decoupling capacitor is required for VDD. It can be located inside the microphone or on the PCB very close to the VDD pin.

Due to high input impedance, care should be taken to remove all flux used during the reflow soldering process.

A 100 Ω resistance is recommended on the clock output of the device driving the FAN3852 to minimize ringing and improve signal integrity.

For optimal PSR, route a trace to the VDD pin. Do not place a VDD plane under the device.

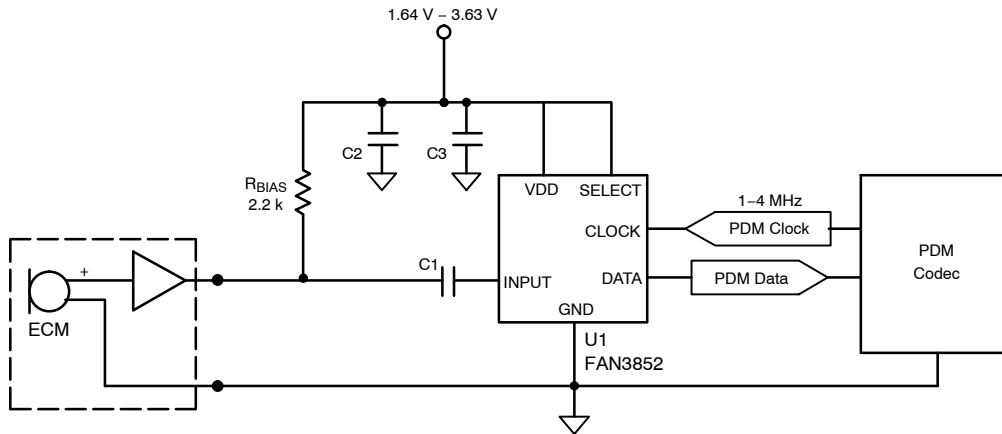
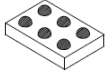


Figure 10. Example Hardware Implementation

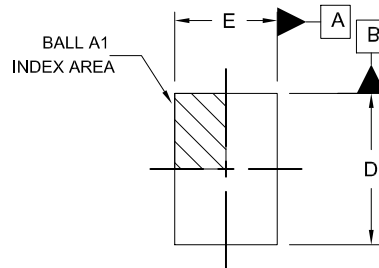
Table 7. RECOMMENDED COMPONENTS

Ref Des	Qty	Description of Options	Package	Manufacturer	Mfg PIN
U1	1	FAN3852 Microphone Pre-Amplifier with Digital Output	WLCSP6	onsemi	FAN3852UC16X
C1	1	Input AC Coupling Capacitor; 1 nF/1000 pF, ≥ 6.3 V, low-leakage	0402	Johansen Dielectrics	500R07W102KV4T
			0402	Murata	GCM155R71H102KA37D
			0603	Taiyo Yuden	UMK107SD102KA-T
C2	1	Primary VDD Decoupling Capacitor; 0.1 μF , MLCC, ≥ 6.3 V	0402	Samsung	CL05B104KO5NNNC
			0402	Yageo	CC0402KRX7R7BB104
			0603	AVX	06033C104KAT4A
C3	1	Optional VDD Decoupling Capacitor; 0.01 μF , MLCC, ≥ 6.3 V	0402	Samsung	CL05B103KB5VPNC
			0402	Murata	GCM155R71H103KA55J
			0603	Yageo	CC0603KRX7R7BB103

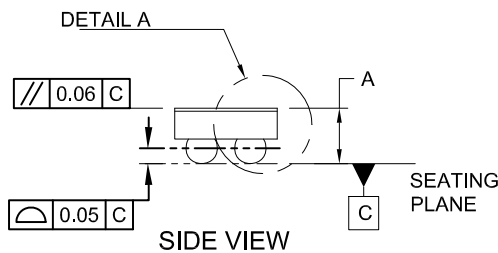


WLCSP6 1.242x0.842x0.457
CASE 567TS
ISSUE A

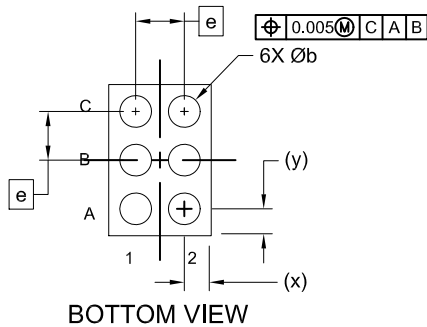
DATE 06 JUN 2019



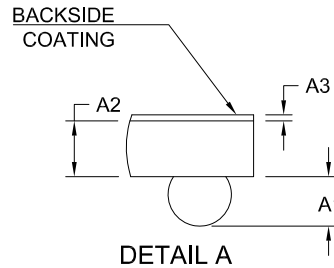
TOP VIEW



SIDE VIEW



BOTTOM VIEW

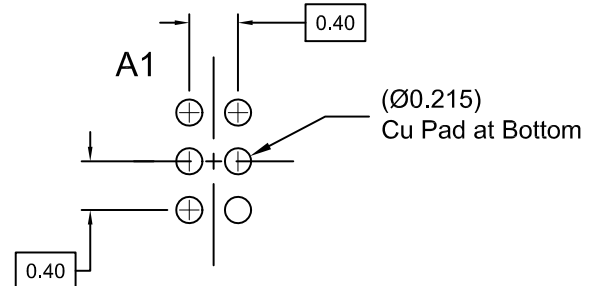


DETAIL A

NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2009.
2. CONTROLLING DIMENSION: MILLIMETERS
3. DATUM C APPLIES TO THE SPHERICAL CROWN OF THE SOLDER BALLS

DIM	MILLIMETERS		
	MIN.	NOM.	MAX.
A	0.419	0.457	0.495
A1	0.183	0.203	0.223
A2	0.214	0.229	0.244
A3	0.022	0.025	0.028
b	0.24	0.26	0.28
D	1.212	1.242	1.272
E	0.812	0.842	0.872
e	0.40 BSC		
x	0.206	0.221	0.236
y	0.206	0.221	0.236



RECOMMENDED MOUNTING FOOTPRINT*
(NSMD PAD TYPE)

* For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

DOCUMENT NUMBER:	98AON13367G	Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped "CONTROLLED COPY" in red.
DESCRIPTION:	WLCSP6 1.242x0.842x0.457	PAGE 1 OF 1

onsemi and onsemi are trademarks of Semiconductor Components Industries, LLC dba onsemi or its subsidiaries in the United States and/or other countries. onsemi reserves the right to make changes without further notice to any products herein. onsemi makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does onsemi assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. onsemi does not convey any license under its patent rights nor the rights of others.

onsemi, **Onsemi**, and other names, marks, and brands are registered and/or common law trademarks of Semiconductor Components Industries, LLC dba "**onsemi**" or its affiliates and/or subsidiaries in the United States and/or other countries. **onsemi** owns the rights to a number of patents, trademarks, copyrights, trade secrets, and other intellectual property. A listing of **onsemi**'s product/patent coverage may be accessed at www.onsemi.com/site/pdf/Patent-Marking.pdf. **onsemi** reserves the right to make changes at any time to any products or information herein, without notice. The information herein is provided "as-is" and **onsemi** makes no warranty, representation or guarantee regarding the accuracy of the information, product features, availability, functionality, or suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. Buyer is responsible for its products and applications using **onsemi** products, including compliance with all laws, regulations and safety requirements or standards, regardless of any support or applications information provided by **onsemi**. "Typical" parameters which may be provided in **onsemi** data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. **onsemi** does not convey any license under any of its intellectual property rights nor the rights of others. **onsemi** products are not designed, intended, or authorized for use as a critical component in life support systems or any FDA Class 3 medical devices or medical devices with a same or similar classification in a foreign jurisdiction or any devices intended for implantation in the human body. Should Buyer purchase or use **onsemi** products for any such unintended or unauthorized application, Buyer shall indemnify and hold **onsemi** and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that **onsemi** was negligent regarding the design or manufacture of the part. **onsemi** is an Equal Opportunity/Affirmative Action Employer. This literature is subject to all applicable copyright laws and is not for resale in any manner.

ADDITIONAL INFORMATION

TECHNICAL PUBLICATIONS:

Technical Library: www.onsemi.com/design/resources/technical-documentation
onsemi Website: www.onsemi.com

ONLINE SUPPORT: www.onsemi.com/support

For additional information, please contact your local Sales Representative at
www.onsemi.com/support/sales